

- 7.1 Constant field scaling allows us represent changes in the scale of CMOS devices with a unitless quantity S . Assuming ideal constant field scaling (that is, scaling the size of a CMOS device without changing the strength of its internal electric field), we may use Table 7.4 to determine that gate delay τ scales by a factor of $\frac{1}{S}$. Thus, using an initial gate delay of $\tau = 500$ ps, we may assume that by scaling the device in question by a factor of $S = 2$, the new gate delay $\tau = \frac{500}{2} = 250$ ps.

Using Table 7.5, we see that repeated wire RC delay per unit length, assuming constant field scaling of gates, scales by a factor of $\sqrt{S}$; in our case, this causes the delay to *increase*, so that $t_{wr} = 500\sqrt{2} \approx 707$ ps. This concords with our intuitive sense of wire scaling and its impact on delay; reduced height in an interconnect will lower its cross-sectional area and increase its resistance, leading to higher delay.

In total, this leads to an overall reduction in delay from $500 + 500 = 1000$ ps to $250 + 707 = 957$ ps, or a 4.3% reduction in delay.

(a)